

Elaborate Quantum Machine Learning Frameworks and Adaptive Feynman Path Integrals for Demyelinating Disease Mitigation in DBS

A Theoretical Nature Preprint

Abstract: We introduce a rigorously elaborated computational framework combining Quantum Machine Learning (QML) and finite mathematical models to calculate and maximize neural recovery in Multiple Sclerosis (MS), alongside an adaptive Deep Brain Stimulation (DBS) target protocol designed to ablate Rosenthal fibers in Alexander's Disease. Employing higher-order finite difference formulations on astrocytic and myelin manifolds, we optimize plaque dissipation via a dynamically tunable cost function. Adaptive ablation is mathematically bounded by discretized Feynman path integrals mapping thermal-electric energy trajectories, avoiding collateral necrosis. This elaborate formulation bridges quantum neural networks with clinical neuro-electrodynamics.

1. Elaborate Finite Mathematics for QML MS Plaque Mitigation

Volumetric plaque density P_n at discrete time step n under DBS electromagnetic field influence is modeled using a second-order Crank-Nicolson finite difference scheme. The QML optimization relies on minimizing a finite state cost function $C(\theta)$:

$$P_{n+1} = P_n - \Delta t \left[\alpha \sum_{k=1}^K \nabla_{\theta_k} C(\theta) \cdot f(E_n, B_n) + \frac{1}{2} D \nabla^2 P_n \right]$$

where D represents the spatial diffusion tensor for remyelination agents. The neural recovery index R_n iterates via a coupled non-linear finite difference equation, mapping recovery as an inverse logistic saturation of plaque reduction:

$$R_{n+1} = R_n + \frac{\gamma \Delta t}{1 + e^{\lambda(P_{n+1} - P_{crit})}} - \beta R_n$$

This discrete mathematical model forms the predictive basis for the MS mitigation tab, accelerating recovery metrics through exact quantum state descent bounding.

2. Elaborate Feynman Path Integrals in Adaptive Ablation

To trace the precise electromagnetic trajectory for ablating Rosenthal fibers, we construct an adaptive grid. The quantum energy propagation amplitude $K(x_f, t_f; x_i, t_i)$ across astrocytic finite volumes is mapped using a discretized Feynman Path Integral sum over all possible conduction paths x_k :

$$K \approx \prod_{j=1}^{N-1} \int dx_j \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{N/2} \exp \left(\frac{i}{\hbar} \sum_{j=1}^N \left[\frac{m(x_j - x_{j-1})^2}{2\varepsilon} - V(x_j)\varepsilon \right] \right)$$

where $\varepsilon = \Delta t$. The potential $V(x_j)$ maps the impedance density of mutant GFAP aggregates. As the path sum interferes destructively outside the Rosenthal fiber, ablation is localized. The cumulative ablation density A_n follows:

$$A_{n+1} = A_n + \kappa \int_{\Omega} |K|^2 \cdot [1 - \rho_{RF}(x)] d^3x$$